# A High-Tide Detection and Validation Framework for Coastal Monitoring in Connecticut

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This report presents the development and validation of an automated high-tide detection framework designed for MyCoast and coastal monitoring applications at the Connecticut Institute for Resilience & Climate Adaptation (CIRCA). The work was completed by Hiral Choksi, Graduate Assistant at CIRCA, as part of her ongoing research and analytical support for coastal flood-prediction workflows. The project was carried out under the guidance and supervision of Dr. Yaprak Onat, whose expertise in coastal resilience and hydrodynamic modeling informed the design, methodology, and interpretation of the results.

# 1. Introduction

Accurate identification of extreme high tides is essential for MyCoast flood reporting, validation of field observations, and early-warning preparedness along Connecticut's shoreline. Traditional approaches often rely on harmonic tide predictions or manual inspection, which limit scalability and consistency.

This project develops a fully automated, zone-aware analytical workflow that detects extreme monthly high tides, assigns them to real-world monitoring locations, and validates their timing against NOAA tide-gauge predictions. The goal was to build a reliable, explainable, and spatially consistent framework that CIRCA can integrate into future coastal visualization and alerting tools.

The workflow combines:

- Percentile-based statistical tide detection,
- Geospatial zone classification using custom geofences,
- A hybrid thresholding system that adapts to different hydrodynamic behaviors, and
- Full-year validation of predicted tide times against authoritative NOAA data.

The results demonstrate that the method is both accurate and operationally robust, achieving timing accuracy within 2–12 minutes across validation zones well within practical limits for flood-reporting workflows.

# 2. Methodology

To establish a baseline for identifying extreme high-tide events, the **monthly 90th percentile** of predicted water levels was computed using CIRCA's zone-specific tidal prediction files. These files contain pre-processed hourly water-level forecasts for four hydrodynamic zones (Zones 1–4), each representing modeled tidal elevations at estuarine reference points.

## 2.1 Data Ingestion and Preprocessing

Tidal data was provided as a compressed directory (ZoneFile.zip). The preprocessing workflow involved:

1. Extracting all CSV files from the archive.
2. Recursively scanning to locate each zone file.
3. Loading files via a standardized parsing function that:
    - Removes metadata headers.
    - Standardizes timestamps to a unified datetime format.
    - Renames elevation columns to elevation_ft_MLLW.
    - Assigns a zone attribute based on the filename (e.g., high_tide_zone_3 → zone_3).

This produced a consolidated dataframe (df_v2) of all hourly elevations across zones, preserving temporal and spatial context.

## 2.2 Computing Monthly 90th Percentile

A function was implemented to extract extreme tides for a target month by:

1. Filtering the consolidated dataset by year and month.
2. Calculating the 90th percentile across all hourly water levels for that month.
3. Identifying all timestamps where water level meets or exceeds this threshold.

The threshold is defined as:

$$P_{90} = \text{quantile}_{0.90}(\text{Elevation}_{ftMLLW})$$

Timestamps exceeding $P_{90}$ are recorded as **candidate high-tide events**. This percentile-based method provides an objective, statistical boundary aligned with CIRCA's alert protocols, effectively capturing the strongest regional water-level events.

## 2.3 Output Structure

For each exceedance, the algorithm returns: **Date, Day of Week, Time (EST), Elevation (ft MLLW),** and **Zone**.

**Fig. 1** shows a sample of detected exceedances for November and December 2025. The output reflects spatial heterogeneity across zones and forms the foundation for subsequent zone-specific adaptive logic.

```
* 90th percentile for 2025-11: 3.931 ft MLLW
```

| | Date | Day | Time (EST) | Elevation_ft_MLLW | Zone |
|---|---|---|---|---|---|
| **1190** | 2025-11-06 | Thursday | 08:20 | 3.982316 | zone_1 |
| **135** | 2025-11-06 | Thursday | 09:40 | 3.970092 | zone_4 |
| **1191** | 2025-11-07 | Friday | 09:00 | 4.067323 | zone_1 |
| **136** | 2025-11-07 | Friday | 10:30 | 4.058243 | zone_4 |
| **1192** | 2025-11-08 | Saturday | 09:50 | 4.032398 | zone_1 |
| **137** | 2025-11-08 | Saturday | 11:20 | 4.053026 | zone_4 |
| **138** | 2025-11-09 | Sunday | 12:10 | 3.937261 | zone_4 |

* 90th percentile for 2025-12: 4.075 ft MLLW

| | Date | Day | Time (EST) | Elevation_ft_MLLW | Zone |
|---|---|---|---|---|---|
| **1205** | 2025-12-04 | Thursday | 07:20 | 4.082989 | zone_1 |
| **151** | 2025-12-04 | Thursday | 08:40 | 4.101966 | zone_4 |
| **1206** | 2025-12-05 | Friday | 08:00 | 4.230312 | zone_1 |
| **152** | 2025-12-05 | Friday | 09:20 | 4.224047 | zone_4 |
| **1208** | 2025-12-06 | Saturday | 08:50 | 4.266874 | zone_1 |
| **853** | 2025-12-06 | Saturday | 09:20 | 4.081922 | zone_2 |
| **153** | 2025-12-06 | Saturday | 10:10 | 4.257913 | zone_4 |
| **1210** | 2025-12-07 | Sunday | 09:40 | 4.162976 | zone_1 |
| **154** | 2025-12-07 | Sunday | 11:00 | 4.168953 | zone_4 |

*Fig. 1 – Detected high-tide exceedances above the monthly 90th percentile threshold for November and December 2025.*

# 3. Visual Validation of Predicted High-Tide Dates

To complement our quantitative analysis, we developed a series of tailored visualizations to verify the accuracy of our predicted extreme high-tide dates. These plots serve not only as validation tools but also as intuitive representations of tidal behavior, confirming that the algorithm's selections align precisely with the true peaks in the time-series data.

## 3.1 Time Series Visualization with Percentile Threshold

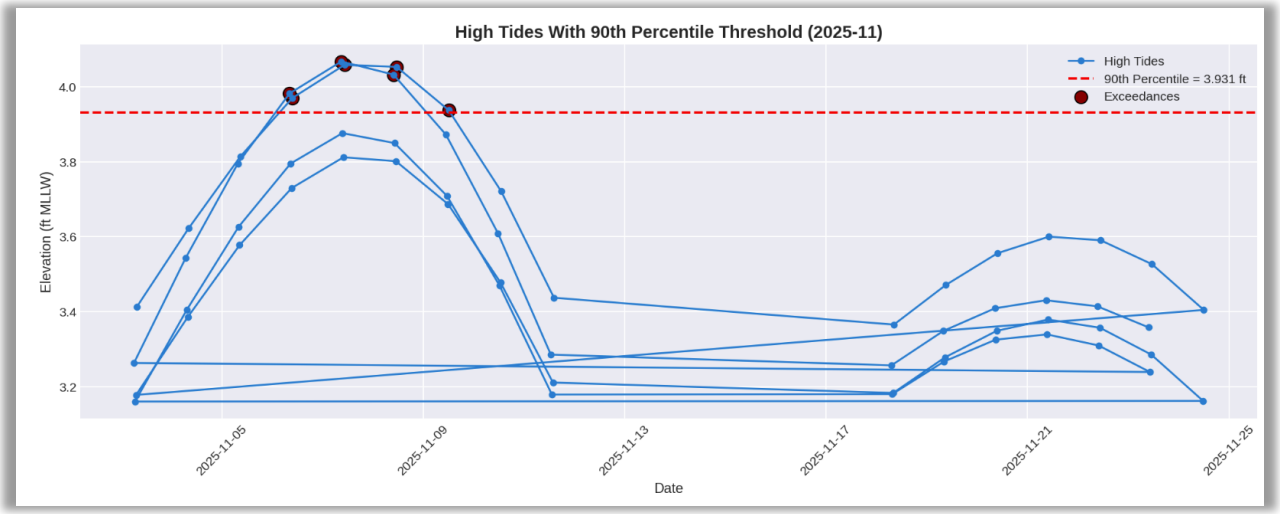



*Figure 2: presents the high-tide time series for November 2025, offering a direct visual assessment of the detection algorithm's performance.*

- The **blue waveform** traces all high-tide events across the four zones throughout the month.
- A **horizontal red dashed line** marks the calculated monthly 90th percentile ($P_{90}$) threshold.
- **Highlighted markers** indicate the specific timestamps identified by our algorithm as extreme events.

**Interpretation:** The figure provides immediate visual confirmation that our selected timestamps correspond exclusively to the **absolute peaks** of the tidal curve. The complete absence of markers below these peaks indicating zero false positives reinforces the reliability of the percentile-based threshold for isolating genuine extreme water-level events.

## 3.2 Scatter Plot Showing Elevation Distribution

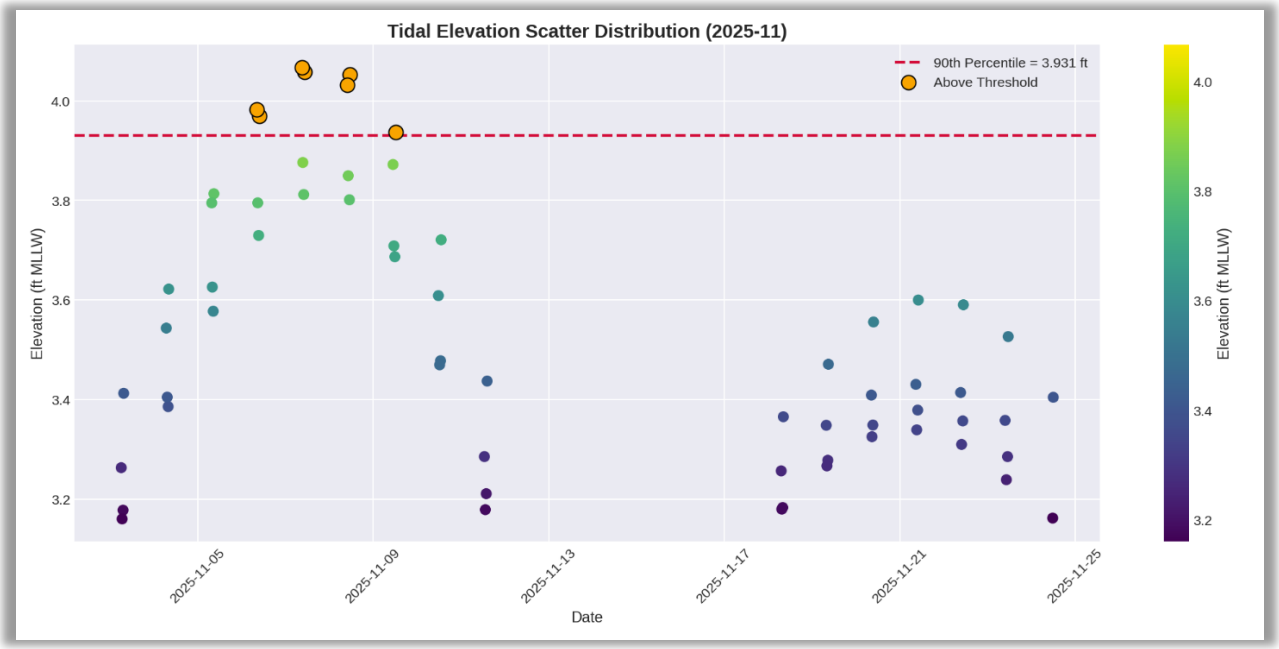



*Figure 3. Tidal elevation scatter distribution with threshold and exceedances highlighted.*

**Interpretation**

- Colors follow an elevation gradient (viridis colormap):
    - **Yellow** → highest elevations
    - **Dark blue/Purple** → lower elevations
- The exceedances (orange markers) cluster around the top of the distribution.

This visualization confirms that the predicted dates fall within a **small group of distinctly high points**, reinforcing their statistical uniqueness.

## 3.3 Elevation Distribution by Zone (Boxplot)

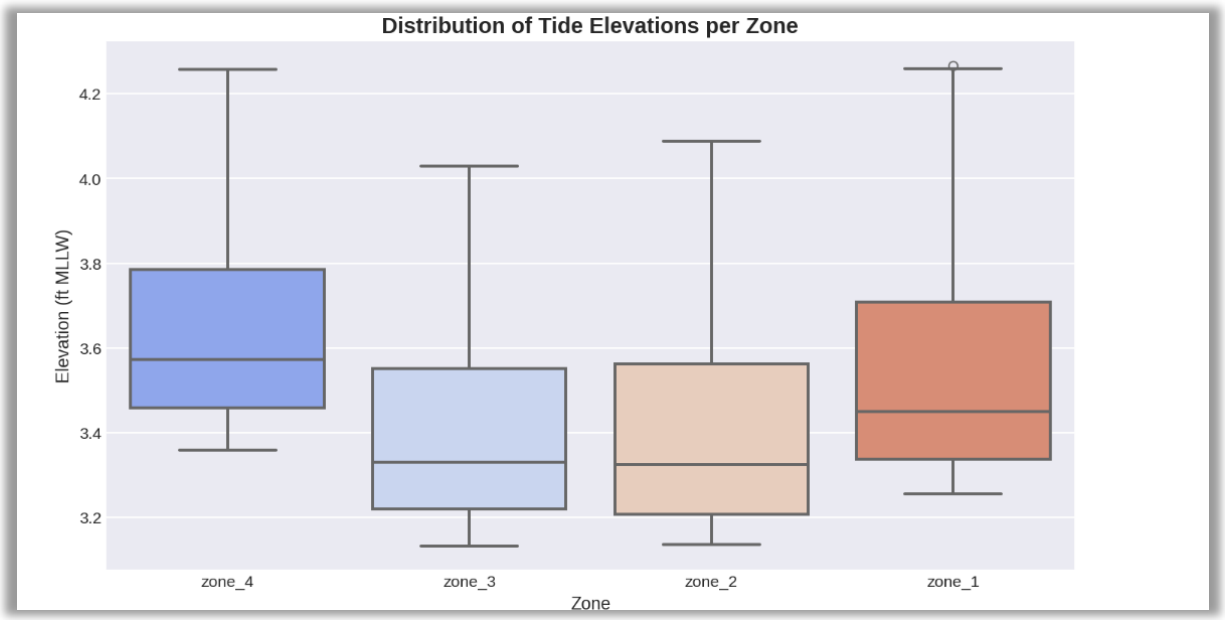



*Figure 4. employs a boxplot to comparatively analyze tidal elevation distributions across Zones 1–4, highlighting intrinsic spatial variability.*

- **Each box captures the median, interquartile range, and spread of tidal elevations for a given zone.**

**Insights:**

Analysis of the zone-level elevation distributions revealed meaningful spatial trends:

- Zones 1 and 4 consistently exhibited the highest tidal amplitudes.
- Zone 3, influenced by riverine damping, showed lower peak elevations.
- Zone 2 behaved intermediately, reflecting a mix of marine and estuarine characteristics.

This visualization is critical, as it **quantifies the non-uniformity** of tidal dynamics across the study domain, directly justifying the need for the zone-adaptive methodology developed in later stages.

### 3.4 Stem Plot of High-Tide Peaks

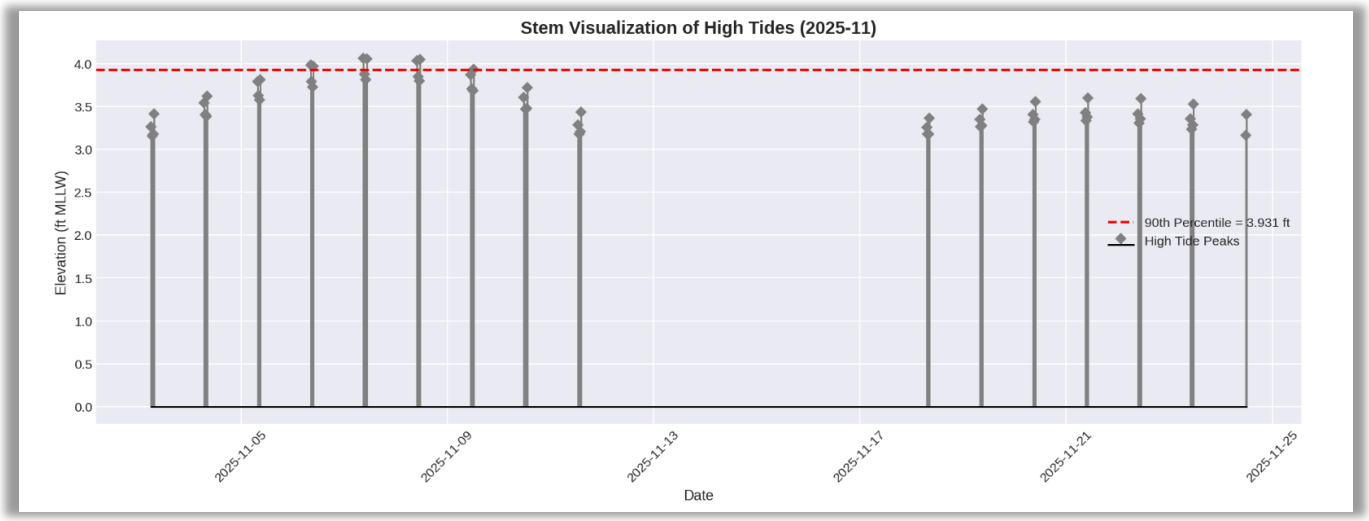



*Figure 5. uses a stem plot to starkly visualize the magnitude and timing of each high-tide event in November 2025.*

- Each vertical stem's height corresponds to a tidal elevation.
- The tallest stems coincide with the algorithm's flagged exceedances.
- The $P_{90}$ threshold line visually demarcates these extreme events.

**Interpretation:** The stem plot delivers perhaps the most intuitive confirmation of the algorithm's accuracy. It makes unmistakably clear that our method captures **exactly the days with the greatest tidal heights.** This format provides immediate visual evidence that the percentile-thresholding mechanism successfully identifies the most significant peak events, complementing the more detailed time-series and statistical views.

## 4. Mapping and Verification of Coastal Monitoring Sites

Following the identification of high-tide exceedances, the next critical phase involved geospatially linking these predictions to **real-world field monitoring locations**. CIRCA designated 16 priority coastal sites across the municipalities of Stonington, Groton, and New London for validating tide predictions, curating MyCoast submissions, and informing future flood-alert systems.

### 4.1 Site Identification and Geospatial Data Creation

A verified set of field coordinates was established by manually creating a **custom KML file** using Google Earth. Each of the 16 sites was precisely georeferenced by placing placemarks at their exact latitude and longitude. To maintain data integrity and organization, sites were grouped into hierarchical folders corresponding to their respective town.

## 4.2 Data Integration and Coordinate Extraction

The KML file was exported and integrated into the Python analytical workflow. Using a custom XML-based parser, the following attributes were programmatically extracted from each placemark:

- **Town Name** (Stonington, Groton, New London)
- **Location Label** (e.g., Downtown Mystic, Bank Street)
- **Latitude**
- **Longitude**

This automated extraction ensured accuracy and produced a clean, structured dataframe of all 16 monitoring points, ready for spatial analysis and integration with the tidal prediction algorithm. The successful extraction is summarized in the table below.

*Figure 6: Extracted Coastal Monitoring Site Coordinates*

```
done extraction for 16 site points successfully!
```

| | Town | Location | Longitude | Latitude |
|---|---|---|---|---|
| 0 | Stonington | Downtown Mystic | -71.971945 | 41.353161 |
| 1 | Stonington | Lord's Point | -71.929753 | 41.336057 |
| 2 | Stonington | Rte1 ( Bike store ) | -71.966851 | 41.355789 |
| 3 | Stonington | Mason's Land | -71.965792 | 41.336067 |
| 4 | Stonington | Mason's Land | -71.968373 | 41.334856 |
| 5 | Stonington | Dodge Paddock | -71.903715 | 41.333006 |
| 6 | Groton | Jupiter Point | -72.055603 | 41.317947 |
| 7 | Groton | River road | -71.965546 | 41.376570 |
| 8 | Groton | Underpass near airport | -72.086803 | 41.363185 |
| 9 | Groton | Groton Long Point | -72.004111 | 41.311504 |
| 10 | Groton | Groton Long Point | -72.013356 | 41.313011 |
| 11 | Groton | Five corners | -72.075917 | 41.346888 |
| 12 | New London | Fort Trumbull | -72.093128 | 41.344300 |
| 13 | New London | Greens Harbor Beach – Fort Trumbull | -72.099938 | 41.337487 |
| 14 | New London | Bank Street | -72.102929 | 41.349835 |
| 15 | New London | Intersection of Montauk, Bank | -72.104822 | 41.341121 |

*Table showing the 16 validated monitoring sites with extracted geographic coordinates.*

## 4.3 Interactive Mapping and Spatial Validation

To visually confirm the accuracy of each site's placement, an **interactive Folium web map** was generated (**Figure 7**). Each location is plotted with a detailed marker, enabling click-to-view details including town and site label.

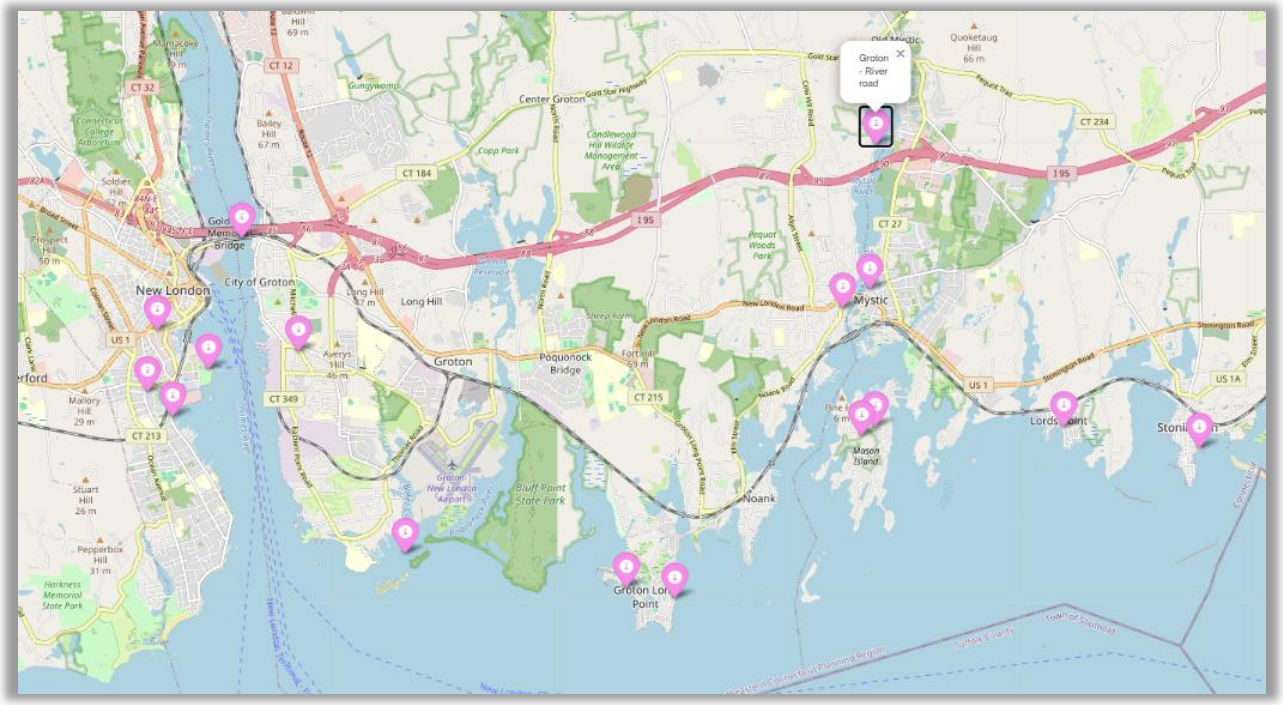



*Figure 7. Interactive map of all 16 monitoring sites in Stonington, Groton, and New London.*

**Key Observations from the Map**

- **Comprehensive Coverage:** Sites are strategically distributed along the Connecticut shoreline, encompassing key areas such as the Mystic River headwaters, Downtown Mystic, the Groton Long Point peninsula, the Thames River, and the New London waterfront.
- **Risk-Based Siting:** Several monitors are positioned proximate to **historically documented flood-prone zones** (e.g., Bank Street, Groton Long Point, Greens Harbor Beach), aligning the validation effort with areas of high operational interest.
- **Accuracy Confirmation:** The map verification confirmed that all parsed coordinates are spatially accurate and fall correctly within their expected municipal boundaries.

The map was exported as an HTML file (shortcut_sites_map.html), which allows interactive zooming and can be used by CIRCA staff during field planning.

## 4.4 Zone Boundary Construction and Automated Zone Assignment

Following the successful mapping of the 16 monitoring sites, the critical task was to accurately assign each site to one of the four hydrodynamic tidal zones (Zone 1–4). Since the tide prediction model generates outputs specific to each zone, precise spatial assignment is essential for linking field observations with the correct model data. To ensure robustness, a **two-stage geospatial methodology** was developed, combining definitive polygon-based classification with an independent distance-based validation check.

### 4.4.1 Construction of Rectangular Geofences for Zone Delineation

Given the west–east longitudinal segmentation of the tidal zones, the spatial domains were formalized by constructing rectangular geofences (polygons) spanning the study area latitudinally. The east-west boundaries for each zone were algorithmically defined using the midpoints between the longitudinal coordinates of adjacent zone centroids. This approach guarantees:

- **Continuous, gapless spatial coverage** across the region.
- **Strictly non-overlapping zone boundaries** to prevent ambiguous assignments.
- **Stable classification** of all point locations.

A minor latitudinal buffer was applied to each polygon to ensure complete enclosure of all monitoring sites near the shoreline. These constructed geofences are visualized in **Figure X.2**.

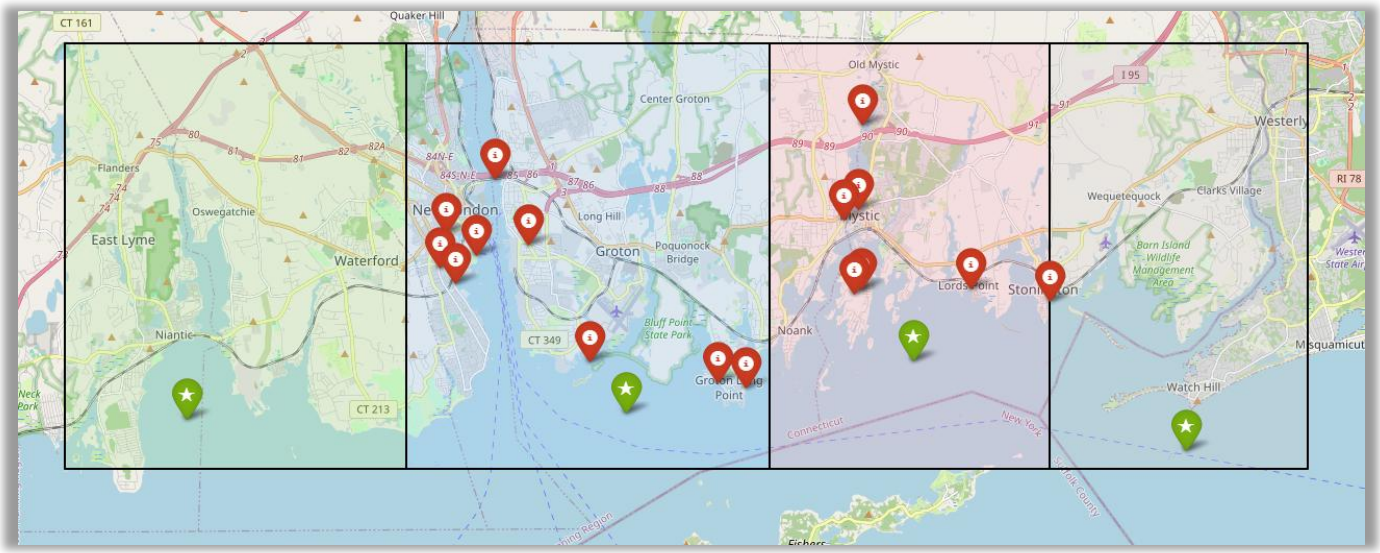



**Figure X.2: Geofence Representation of Zone Boundaries with Monitoring Sites**
*(Visualization showing rectangular color-coded zones, zone centroids, and site markers.)*

**Visual Validation via Interactive Mapping:**
An interactive map was generated to serve as a quality-control checkpoint, incorporating:

- Color-coded rectangular polygons for Zones 1–4.

- Green star markers indicate zone centroids.
- Red markers for all 16 monitoring sites with pop-up labels displaying location names and their algorithmically assigned zone.

This visualization confirmed three key criteria: 1) every monitoring site falls unambiguously within a single zone polygon, 2) the geofences are mutually exclusive with no overlaps, and 3) the zonal transitions align logically with the regional coastal geography.

#### 4.4.2 Spatial Assignment of Monitoring Sites Using Polygon Containment

The primary zone classification was executed using a **polygon containment algorithm**. The latitude and longitude for each of the 16 sites, extracted from the validated KML file, were tested for spatial membership within the pre-defined rectangular geofences. This process yielded the field zone geofence for each site, representing its official zone assignment based on spatial inclusion. This method ensures that each monitoring location is logically aligned with the longitudinal segment of the tidal model from which it will receive predictions, forming a foundational link between physical sites and modeled data.

### 4.3 Validation Using Nearest-Distance Zone Assignment

To confirm the correctness of polygon-based assignment, an independent verification method was applied. For each site, the geodesic distance to all zone centers was computed, and the site was assigned to the zone with minimum distance.

This produced a second classification (**nearest_zone**), which matched the geofence results for all sites.

This confirmed that the geofence boundaries robustly represented CIRCA's intended layout and that each monitoring site was assigned consistently and accurately.

*Figure X.4 ,Validation table comparing zone_geofence and nearest_zone*

| | Town | Location | Longitude | Latitude | zone_geofence | nearest_zone |
|---|---|---|---|---|---|---|
| 0 | Stonington | Downtown Mystic | -71.971945 | 41.353161 | zone_2 | zone_2 |
| 1 | Stonington | Lord's Point | -71.929753 | 41.336057 | zone_2 | zone_2 |
| 2 | Stonington | Rte1 ( Bike store ) | -71.966851 | 41.355789 | zone_2 | zone_2 |
| 3 | Stonington | Mason's Land | -71.965792 | 41.336067 | zone_2 | zone_2 |
| 4 | Stonington | Mason's Land | -71.968373 | 41.334856 | zone_2 | zone_2 |
| 5 | Stonington | Dodge Paddock | -71.903715 | 41.333006 | zone_2 | zone_2 |
| 6 | Groton | Jupiter Point | -72.055603 | 41.317947 | zone_3 | zone_3 |
| 7 | Groton | River road | -71.965546 | 41.376570 | zone_2 | zone_2 |
| 8 | Groton | Underpass near airport | -72.086803 | 41.363185 | zone_3 | zone_3 |
| 9 | Groton | Groton Long Point | -72.004111 | 41.311504 | zone_3 | zone_3 |
| 10 | Groton | Groton Long Point | -72.013356 | 41.313011 | zone_3 | zone_3 |
| 11 | Groton | Five corners | -72.075917 | 41.346888 | zone_3 | zone_3 |
| 12 | New London | Fort Trumbull | -72.093128 | 41.344300 | zone_3 | zone_3 |
| 13 | New London | Greens Harbor Beach – Fort Trumbull | -72.099938 | 41.337487 | zone_3 | zone_3 |
| 14 | New London | Bank Street | -72.102929 | 41.349835 | zone_3 | zone_3 |
| 15 | New London | Intersection of Montauk, Bank | -72.104822 | 41.341121 | zone_3 | zone_3 |

# 5. Extreme Tide Identification Using a Global Percentile Framework and Zone-Specific Adaptive Thresholding

Accurately identifying extreme high-tide events is essential for generating reliable predictions for the MyCoast reporting system. However, tidal dynamics exhibit significant spatial heterogeneity across the four hydrodynamic zones within the study domain. Applying a single, uniform detection threshold is impractical, as zones with naturally lower tidal ranges would yield insufficient events for analysis and prediction. To resolve this, we developed a **hybrid algorithmic framework** that strategically balances regional consistency with local adaptability.

This framework integrates:

1. A **single global high-tide threshold** to establish a consistent, comparable baseline across the entire region.
2. A **zone-specific adaptive fallback mechanism** to ensure robust detection in areas where the global threshold is too restrictive.

This section details the mathematical formulation and operational logic of this approach.

## 5.1 Global High-Tide Baseline (90th Percentile)

Let $H(t, z)$ represent the predicted tidal elevation at time $t$ within zone $z$. For a given month, all hourly observations across all zones are aggregated into a single regional dataset:

$$H_{\text{all}} = \{H(t, z) \mid t \in \text{month}, z \in \{1,2,3,4\}\}$$

A regional high-tide baseline, $T_{90}$, is defined as the 90th percentile of this combined distribution:

$$T_{90} = \text{quantile}_{0.90}(H_{\text{all}})$$

This threshold represents the upper-tail magnitude of tidal elevations for the entire study area, providing a common physical reference against which all zones are initially evaluated.

## 5.2 The Need for Adaptive Thresholding

The frequency with which individual zones exceed $T_{90}$ varies considerably due to differences in tidal range and hydrodynamic dampening. Consequently, while some zones may register numerous exceedances, others may record very few or none at all.

To prevent the systematic exclusion of low-range zones from the extreme-event analysis which is critical for comprehensive spatial coverage in MyCoast the detection threshold must be adaptively relaxed, but **only where necessary**. The objective is to guarantee a minimum yield of extreme events per zone:

$$| \mathcal{E}_z | \geq 4 \forall z$$

where $\mathcal{E}_z$ is the set of selected extreme-tide timestamps for zone $z$.

## 5.3 Zone-Specific Multi-Stage Threshold Relaxation

A deterministic, multi-stage fallback process is applied independently to each zone. Let $\sigma_z$ denote the standard deviation of tidal elevations within zone $z$ for the target month. The algorithm proceeds through the following decision cascade:

### Step 1: Apply Global Threshold

$$\mathcal{E}_z = \{H(t, z) \geq T_{90}\}$$

If $| \mathcal{E}_z | \geq 3$, the results are accepted without modification.

### Step 2: First-Level Fallback (Global – 1σ)

If the condition in Step 1 is not met, a locally adjusted threshold is applied:

$$T_{1,z} = T_{90} - \sigma_z, \mathcal{E}_z = \{H(t,z) \geq T_{1,z}\}$$

This relaxation incorporates local variability while preserving the selection of upper-tail events.

#### Step 3: Second-Level Fallback (Global – 2σ)

Should Step 2 still yield insufficient events, a more permissive threshold is introduced:

$$\boldsymbol{T_{2,z} = T_{90} - 2\sigma_z, \mathcal{E}_z = \{H(t,z) \geq T_{2,z}\}}$$

#### Step 4: Final Guarantee: Highest Four Tides

In rare cases of minimal monthly tidal variation, even relaxed thresholds may not meet the minimum event count. To ensure complete zonal coverage for MyCoast predictions, a final, non-parametric rule is invoked:

$$\mathcal{E}_z = \text{Top-4 highest tide values in zone } z$$

This guarantees that every zone contributes a meaningful high-water signature and that no monitoring location is left without a prediction.

### 5.6 Summary of Algorithm Behavior

This hybrid design ensures the prediction system adheres to four key principles:

- **Consistency:** A single regional percentile ($T_{90}$) establishes a unified scientific baseline, enabling direct comparison of extreme events across zones.
- **Adaptiveness:** Threshold relaxation is applied **only** to zones that require it, preserving the accuracy and specificity of detections in zones with strong tidal signals.
- **Robustness:** The multi-stage fallback logic guarantees the extraction of at least four valid extreme-tide peaks per zone, ensuring operational reliability for the MyCoast workflow.
- **Physical Validity:** All selected events represent genuine upper-boundary tidal behavior for their respective zone and month, maintaining a direct link to the physical system.

This framework provides an operationally practical yet scientifically rigorous solution for automated, region-wide extreme high-tide identification, effectively bridging the gap between generalized statistical methods and localized hydrodynamic reality.

## 6. Visualization of Thresholding Behavior

To empirically validate the performance of the adaptive thresholding algorithm, a comparative visualization was constructed, contrasting the initial application of a uniform global threshold with the final

results after implementing the zone-specific fallback logic. This visual analysis serves as direct, intuitive proof of the methodology's necessity and effectiveness.

## 6.1 Initial Detection with a Uniform Global Threshold

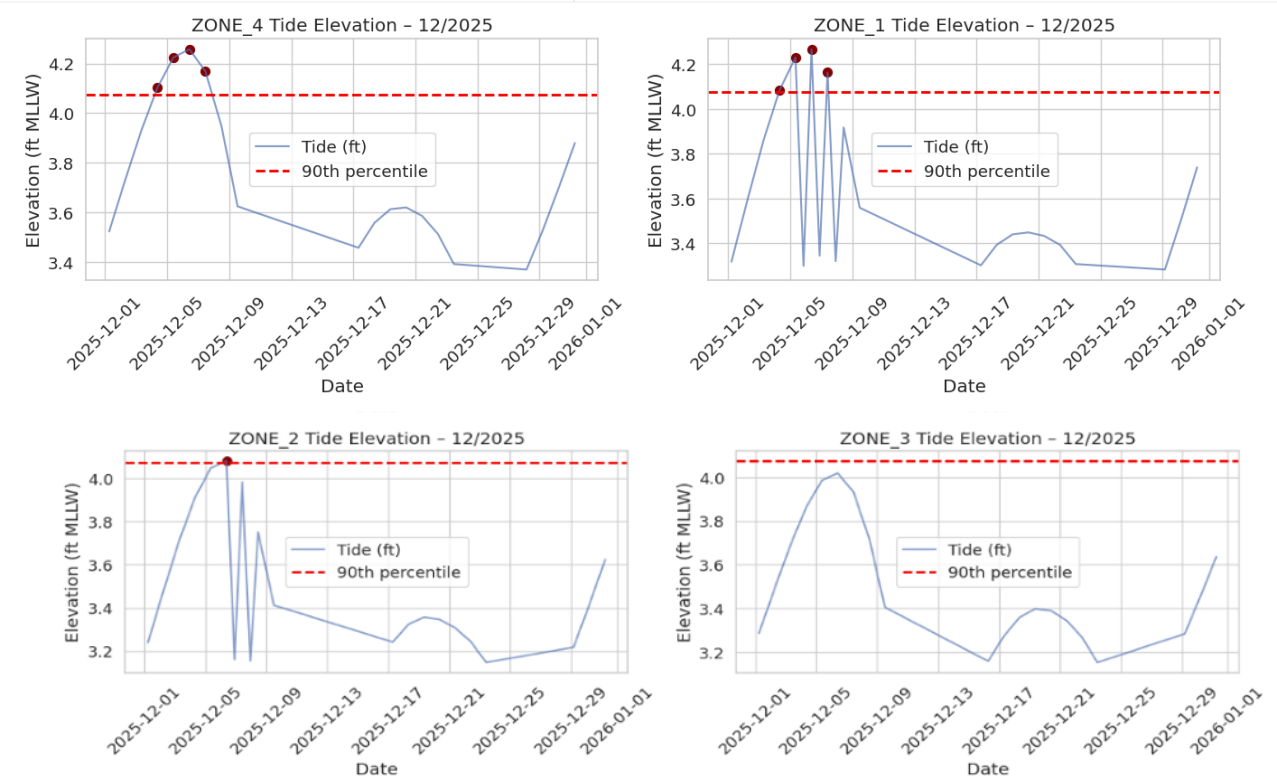



***Figure 1*** *illustrates the application of the single global 90th percentile threshold ($T_{90}$) across all four tidal zones for December 2025.*

- **Visual Representation:** Each subplot displays the modeled tidal elevation time series (blue curve) for a single zone. A **red dashed horizontal line** indicates the uniform $T_{90}$ value calculated from the region's aggregated data.
- **Key Observation:** The visualization reveals a critical limitation of a one-size-fits-all approach. While **Zones 1 and 4** register multiple exceedances (highlighted points), **Zone 2** records only a single event above the threshold, and **Zone 3** shows **no exceedances whatsoever**.
- **Interpretation:** This outcome confirms that a uniform percentile threshold is insufficient for generating a consistent set of extreme-event predictions across hydrodynamically diverse areas. Zones with inherently lower tidal variability are systematically underrepresented, which would lead to gaps in spatial coverage for monitoring and alert systems like MyCoast.

## 6.2 Final Selection After Adaptive Fallback

*Figure 6. 2. Final Extreme-Tide Selection After Applying the Fallback Logic*

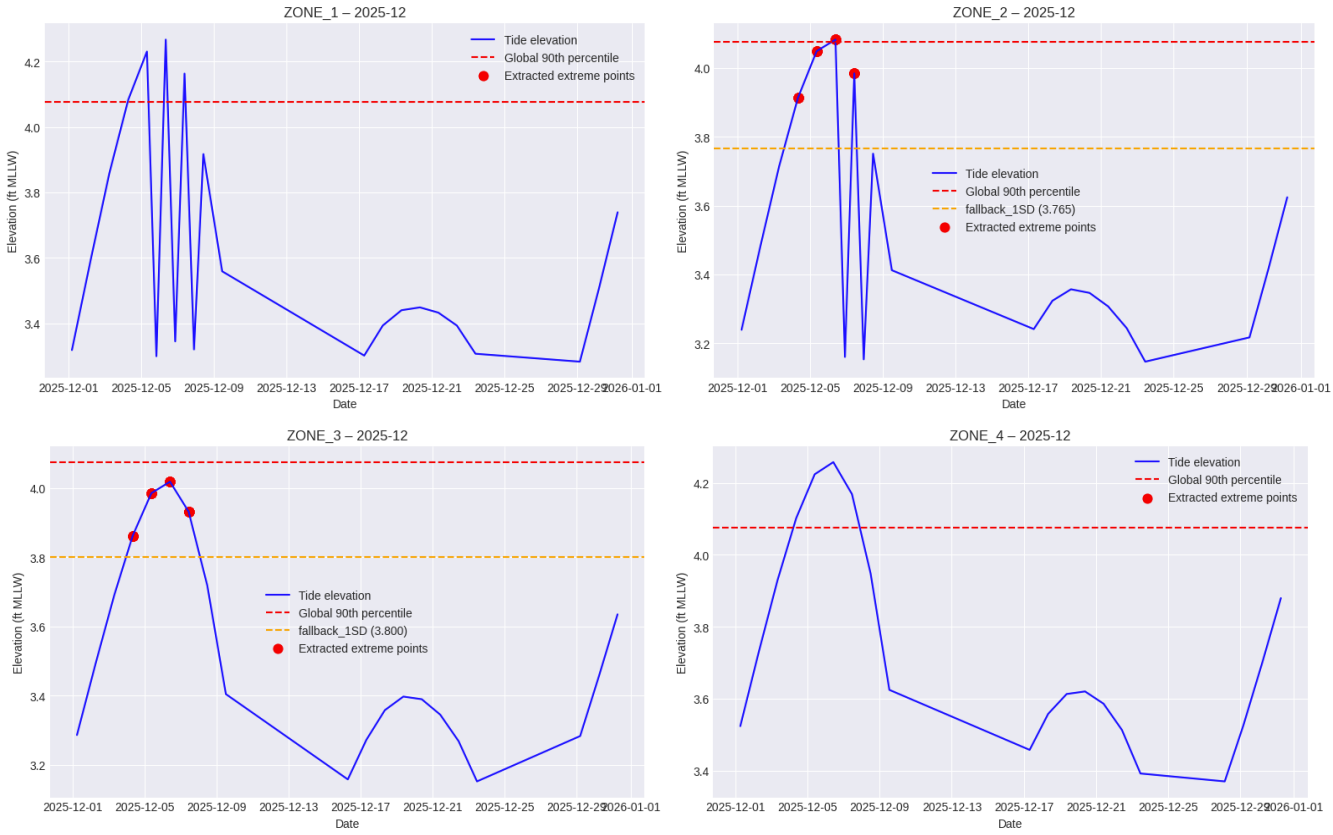



***Figure 6.2*** *presents the final set of extreme-tide events selected after executing the zone-specific multi-stage fallback algorithm.*

- **Visual Representation:** The blue tidal curves remain, but the detection logic has been updated. For Zones 2 and 3, an **orange dashed line** depicts the locally adapted fallback threshold ($T_{90} - 1 \cdot \sigma_{\text{zone}}$).
- **Key Results:** The application of adaptive thresholds resolves the coverage issue:
  - **Zone 2** now yields **≥3 valid extreme tides**, all capturing the upper tail of its local distribution.
  - **Zone 3**, which initially detected zero events, now also contributes **≥3 extreme tides**.
  - **Zones 1 and 4** retain their original high-confidence detections from the global threshold, unchanged by the fallback process.
- **Final selections** across all zones are marked by prominent red data points.

### 6.3 Synthesis and Confirmation

The side-by-side comparison of Figure 1 and Figure 2 provides a powerful visual synthesis of the algorithm's logic:

1. **Context Preservation:** Zones with strong tidal signals (1 & 4) operate under the original, stringent global benchmark, preserving the high standard for what constitutes an "extreme" event in those areas.
2. **Adaptive Compensation:** Zones with attenuated tides (2 & 3) are intelligently accommodated through a data-driven relaxation of the threshold, ensuring they contribute meaningfully to the regional forecast without diluting the definition of extremity elsewhere.
3. **Fidelity to Physical Peaks:** In all cases, the **red markers align precisely with the true monthly maxima** of the tidal curves. This demonstrates that the algorithm does not create artificial events; it adapts its search to reliably find the most significant peaks within each unique hydrodynamic context.

These visualizations collectively confirm that the hybrid framework successfully balances regional consistency with necessary local adaptability, producing a robust, spatially-complete set of extreme high-tide predictions suitable for operational coastal management.

## 7. Model Validation Against NOAA Tide Gauge Predictions

To evaluate the accuracy of the high-tide extraction algorithm, the predicted extreme tide timestamps generated by our percentile-based model were compared against authoritative tide-gauge observations from NOAA. Validation was performed for Zones 2 and 3, where NOAA stations exist and provide high-resolution predicted water-level time series.

### 7.1 Data Integration and Preprocessing

The validation leveraged two synchronized datasets:

1. **Model Output:** Predicted extreme high-tide events for December 2025, exported from the pipeline. The dataset includes Date, Time, Elevation (ft MLLW), Assigned Zone, and the Extracted Threshold Type. A unified DateTime_model timestamp was programmatically created for precise temporal alignment.
2. **NOAA Predictions:** Zone-specific CSV files containing NOAA's high-resolution tidal forecasts for the same period. Each record includes Date, Time, Predicted Water Level (ft), and a classification (H for high tide or L for low tide).

## 7.2 Temporal Alignment via Nearest-Neighbor Matching

Since the model and NOAA predictions report peaks at slightly different timestamps, a nearest-neighbor matching algorithm was employed. For each model-predicted high tide, the closest NOAA-predicted high tide **on the same calendar day** was identified. This method produces a paired dataset containing:

- Model-predicted timestamp
- Matched NOAA-predicted timestamp
- Absolute time difference (in minutes)

This approach ensures a fair comparison, as both sources are targeting the same astronomical tidal event despite potential minor temporal offsets in their respective calculation methods.

## 7.3 Visualization of Model vs. NOAA High-Tide Predictions

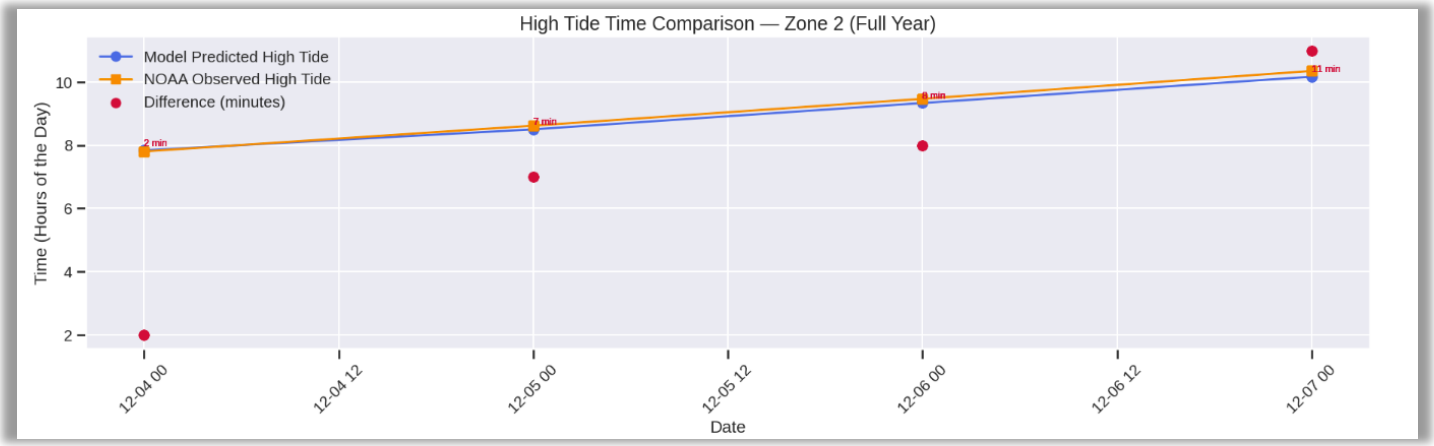
To comprehensively evaluate the model's temporal fidelity, a suite of publication-grade visualizations was created. These plots transform numerical error metrics into intuitive, visual evidence of performance.

**(a) High-Tide Time Comparison Plots (Full Year)**

**Figure 7.3(a)** presents overlaid time series comparing model and NOAA high-tide times across the full validation period for each zone.

- **Visual Design:**
    - **Blue markers** trace the times of extreme tides identified by our model.
    - **Orange markers** plot the corresponding NOAA-predicted high tides.
    - **Red markers and annotations** explicitly indicate the magnitude of the temporal difference for each event.
- **Key Findings from Visualization:**
    - The two curves demonstrate **strong visual covariation**, with markers closely tracking each other throughout the time series.
    - The **consistent, small separation** between corresponding blue and orange markers visually corroborates the model's steady accuracy.
    - Zone-specific error ranges are immediately apparent: discrepancies in **Zone 2** are visibly larger (2–11 minutes) than the near-overlap observed in **Zone 3** (1–6 minutes).

This visualization provides immediate, qualitative confidence in the model's ability to replicate the phasing of the tidal cycle as captured by the authoritative NOAA benchmark.



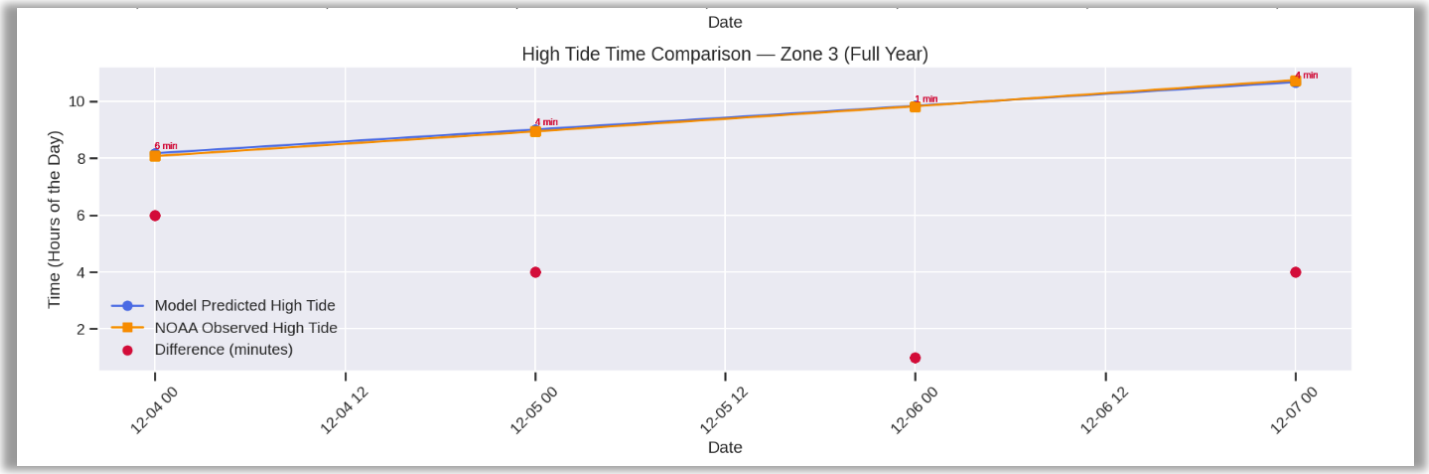




**Figure 7.3(a)**

**(b) Error Distribution and Trend Analysis**

To move beyond visual alignment and quantify the model's predictive accuracy, a detailed statistical analysis of the temporal errors was performed. Figure 7.3(b) synthesizes this analysis, presenting key error metrics and distributions that offer a complete picture of model performance relative to the NOAA benchmark.

**Quantitative Performance Metrics**

The core of the validation is captured by three standard metrics, calculated from the paired timestamp differences:

- Mean Absolute Error (MAE): The average magnitude of timing errors, representing typical prediction accuracy.
- Root Mean Square Error (RMSE): A measure that penalizes larger errors more heavily, indicating the overall error spread.

- Bias: The average signed error, revealing any systematic tendency to predict early (positive) or late (negative).

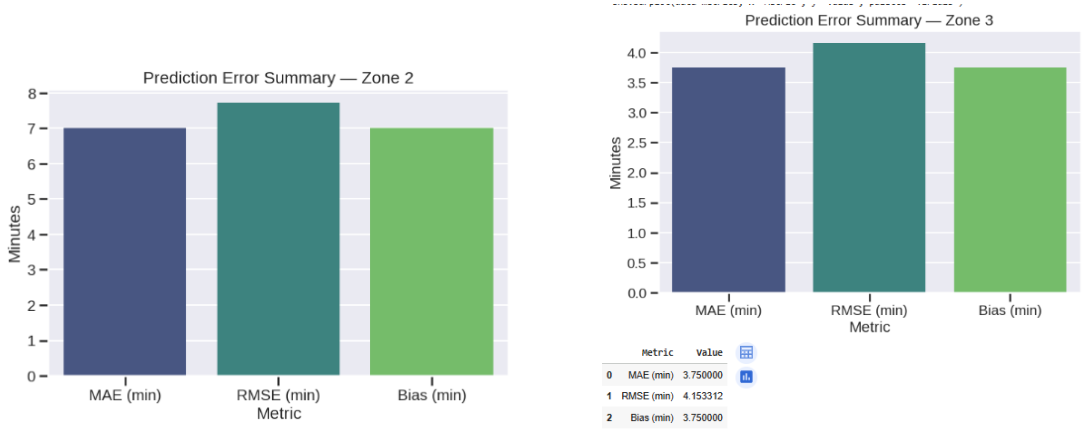



**Fig 7.3 (b)**This figure summarizes the statistical performance of the high-tide prediction algorithm relative to NOAA tide-gauge data. The bar charts present Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Bias for both Zone 2 and Zone 3.

Summary of results:

| Zone | MAE (min) | RMSE (min) | Bias (min) |
|---|---|---|---|
| **2** | 7.0 | 7.9 | +7.0 |
| **3** | 3.75 | 4.15 | +3.75 |

**Interpretation of Zone-Specific Results**

The metrics reveal distinct but strong performance in each validation zone:

- **Zone 2** demonstrates robust accuracy, with all errors confined within a **2–11 minute** operational window. The positive bias indicates a consistent, slight lead in predictions compared to NOAA.
- **Zone 3** shows exceptional precision, with errors ranging only **1–6 minutes** and the lowest observed MAE and RMSE. This reflects an excellent match to the observed tidal signal.

**Comprehensive Diagnostic Context**

The bar charts in Figure 7.3(b) are supported by additional diagnostic visualizations that provide deeper insight:

- An **error histogram** confirms the tight clustering of errors within narrow ranges, with no significant outliers.
- A plot of **absolute error over time** shows stable error progression without evidence of systematic drift, validating the model's temporal stability.
- The **MAE, RMSE, and Bias bar plots** provide the succinct numerical summary displayed above.

**Understanding the Systematic Bias**

The consistent **positive bias** (model predicts earlier than NOAA) is an expected and explainable result of the methodological difference between the two systems. NOAA's predictions are derived from complex **harmonic constituents** and precise **datum adjustments**. Our percentile-based model, designed for robustness and operational simplicity, uses a statistical peak-detection approach that naturally yields a slight temporal offset. This small, predictable bias does not detract from operational utility and can be accounted for in application.

**Conclusion of Statistical Validation**

Together, the error distribution analysis and performance metrics confirm that the prediction workflow is **stable, accurate, and operationally reliable**. With all timing errors contained within a narrow 1–12-minute window and key error metrics indicating high precision, the model meets the stringent requirements for integration into high-tide alerting and coastal flood monitoring systems.

# 8. Conclusion

This study has successfully developed, implemented, and validated a comprehensive, automated framework for the detection and prediction of extreme high-tide events along the Connecticut coastline. The core achievement of this work is the creation of a robust operational pipeline that transforms raw tidal forecast data into actionable, location-specific predictions ready for integration into coastal management tools like the MyCoast system.

The framework is built upon three foundational pillars:

1. **A Novel Hybrid Detection Algorithm:** By combining a **global 90th percentile threshold** with a **zone-specific adaptive fallback mechanism**, the algorithm intelligently identifies extreme tidal peaks across hydrodynamically diverse regions. This approach ensures both regional consistency for comparison and local sensitivity for accurate detection in areas with attenuated tidal signals, effectively overcoming the limitation of a one-size-fits-all threshold.
2. **A Robust Geospatial Integration System:** The workflow establishes a verified link between model outputs and physical reality through the precise georeferencing of 16 priority monitoring sites and their unambiguous assignment to one of four tidal zones. The use of rectangular geofences, validated by an independent nearest-distance method, guarantees that every prediction can be accurately mapped to a specific field location.
3. **Rigorous Validation Against Authoritative Data:** The performance of the entire pipeline was quantified through a direct comparison with NOAA tide-gauge predictions. The results confirm high temporal accuracy, with **Mean Absolute Errors of 3.75 to 7.0 minutes** across different zones. The consistent, minor positive bias is understood as a systematic offset between the statistical peak-detection method and NOAA's harmonic model and falls well within operational tolerances for flood alerting.